

Maths Chapter 6

Population: Entire group we want to study. Ex: all students in a uni

Parameters: μ = population mean
 σ^2 = population variance

Sample: A smaller subset of the population.

Parameters: $\bar{X}$ = sample mean

S^2 = sample variance

Formulae:

$$\textcircled{1} \quad S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad \text{and} \quad E[S^2] = \sigma^2$$

$$\textcircled{2} \quad \sigma^2 = \frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2$$

$$\textcircled{3} \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

↳ variance of Sample Mean: how much the sample mean changes across multiple samples.

Central Limit Theorem (CLT)

For a set of independent RVs : $X_1, X_2, \dots, X_n$ with the same mean (μ) and variance (σ^2), then for large n , the sum $X_1 + X_2 + \dots + X_n$ is normally distributed even if the RVs themselves aren't normal.

Let $Y = X_1 + X_2 + \dots + X_n$ $Y \sim \text{Gaussian}(n\mu, n\sigma^2)$

$$E[Y] = n\mu$$

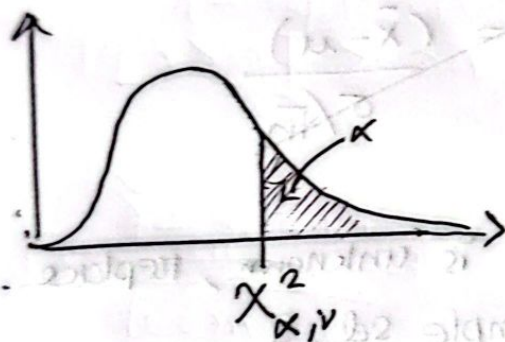
$$\text{Var}[Y] = n\sigma^2, \quad \text{SD}[Y] = \sigma\sqrt{n}$$

$$\text{And } \frac{Y - E[Y]}{\text{SD}[Y]} = \frac{Y - n\mu}{\sigma\sqrt{n}} \approx Z$$

Chi-Square Distribution

For n standard normal distributions, $Z_1, Z_2, \dots, Z_n$, where $Z_i \sim N(0,1)$, X follows a chi-square distribution if

$X = Z_1^2 + Z_2^2 + \dots + Z_n^2$. Thus $X \sim \chi_n^2$, and it has n degrees of freedom.



$$P[X \geq \chi^2_{\alpha, n}] = \alpha$$

↑
critical
value

For a normal population,

$$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$$

t - Distribution

t - distribution is used when σ is unknown and we replace it with S .

Derivation

$$Z = \frac{\sqrt{n} (\bar{X} - \mu)}{\sigma}$$

$$\bar{Z} = \frac{(\bar{X} - \mu)}{\sigma/\sqrt{n}}$$

Since σ is unknown, replace with sample sd, S

$$T_n = \frac{\sqrt{n} (\bar{X} - \mu)}{S}$$

$$T_n = \frac{\sqrt{n} (\bar{X} - \mu)/\sigma}{S/\sigma}$$

$$T_n = \frac{\bar{Z}}{S/\sigma}$$

we know

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

$$\Rightarrow \frac{S}{\sigma} = \sqrt{\frac{\chi^2_{n-1}}{(n-1)}}$$

$$-T_{n-1} = \frac{Z}{\sqrt{\frac{\sum_{i=1}^n x_i^2}{n-1}}}$$

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S} = \frac{Z}{\sqrt{\sum_{i=1}^n x_i^2 / (n-1)}}$$

Chapter 7 : MLE

Parameters:

Ex: $X \sim \text{Poisson}(\lambda)$

↖ parameter

unknown parameters are represented by θ

Estimator: Formula or statistic used to estimate an

unknown parameter. Ex: $\hat{\mu} = \bar{X}$ ↖ estimator

sample : 2, 4, 6

$$\bar{X} = 4$$

↖ estimate

MLE:

Estimation of an unknown parameter to maximise the probability of observed data. Example scenario:

given a set of data: $x_1, x_2, \dots, x_N$ with unknown mean, μ ;

Estimating μ involves maximizing $L(\mu)$ where

$$L(\mu) = f(x_1; \mu) f(x_2; \mu) f(x_3; \mu) \dots f(x_N; \mu)$$

$\uparrow$
probability
density function

$\hat{\mu}$ = value of μ that maximizes $L(\mu)$

NOTE: for Gaussian distribution $\hat{\mu}$ turns out to be:

$$\hat{\mu} = \frac{1}{N} \sum_{n=1}^N x_n$$

MLE of a Bernoulli Parameter

n trials (ind.) each with successful prob. p .

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{otherwise} \end{cases}$$

$$P[X_i = x] = p^{x_i} (1-p)^{1-x_i}$$

$$L(p) = P[X_1 = x_1] \dots P[X_n = x_n]$$

$$L(p) = p^{x_1} (1-p)^{1-x_1} \dots p^{x_n} (1-p)^{1-x_n}$$

$$L(p) = p^{\sum x_i} (1-p)^{n - \sum x_i} \quad \text{where } \sum x_i = \text{no. of succ.}$$

$$\ln L(p) = \sum x_i \ln(p) + (n - \sum x_i) \ln(1-p)$$

$$\Rightarrow \frac{d}{dp} \ln L(p) = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p}$$

Set the derivative to 0 to find $\hat{p}$

$$\Rightarrow \frac{\sum x_i}{\hat{p}} = \frac{n - \sum x_i}{1 - \hat{p}}$$

$$\Rightarrow \hat{p} (n - \sum x_i) = (1 - \hat{p}) \sum x_i$$

$$\Rightarrow \sum x_i = n \hat{p}$$

$\Rightarrow$

$$\boxed{\hat{p} = \frac{\sum_{i=1}^n x_i}{n}}$$

MLE of a Poisson Parameter

$$P[X=x] = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$L(\lambda) = P[X_1=x_1] \dots P[X_n=x_n]$$

$$L(\lambda) = \frac{e^{-\lambda} \lambda^{x_1}}{x_1!} \dots \frac{e^{-\lambda} \lambda^{x_n}}{x_n!}$$

$$= \frac{e^{-n\lambda} \lambda^{\sum x_i}}{x_1! \dots x_n!}$$

$$\ln(L(\lambda)) = -n\lambda \ln(e) + \sum x_i \ln(\lambda) - \ln(x_1! \dots x_n!)$$

$$\ln(L(\lambda)) = -n\lambda + \sum x_i \ln(\lambda) - \ln\left(\prod_{i=1}^n x_i!\right)$$

$$\frac{d}{d\lambda} \ln(L(\lambda)) = -n + \frac{\sum x_i}{\lambda} = 0$$

$$\Rightarrow -n + \frac{\sum x_i}{\hat{\lambda}} = 0$$

$$\Rightarrow \frac{\sum x_i}{\hat{\lambda}} = n$$

$$\Rightarrow \boxed{\hat{\lambda} = \frac{\sum x_i}{n}}$$

Ex: 7.2d

for a poisson var,

$$\hat{\lambda} = \frac{\sum x_i}{n} = 2.7$$

$$X \sim \text{Poisson}(2.7)$$

$$P[X \leq 2] = \sum_{x=0}^{x=2} \frac{e^{-2.7} (2.7)^x}{x!}$$

$$= 0.4936 \text{ (Ans)}$$

MLE of a Normal Population with 2 unknown parameters

$X_i \sim N(\mu, \sigma^2)$, both μ and σ^2
are unknown

$$f(x_i) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right]$$

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(x_i - \mu)^2}{2\sigma^2}\right]$$

$$L(\mu, \sigma^2) = \left(\frac{1}{2\pi}\right)^{\frac{n}{2}} \times \frac{1}{\sigma^n} \times \exp\left[-\frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2}\right]$$

$$\ln(L(\mu, \sigma^2)) = \frac{n}{2} \ln\left(\frac{1}{2\pi}\right) - n \ln(\sigma)$$

$$- \frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2} \ln(e)$$

$$\Rightarrow \ln(L(\mu, \sigma^2)) = -\frac{n}{2} \ln(2\pi) - n \ln(\sigma)$$

$$- \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

First diff wrt μ ,

$$\frac{d}{d\mu} (l(\mu, \sigma^2)) = \frac{2}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

$$= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

$$= \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \hat{\mu}) = 0$$

$$\Rightarrow \sum_{i=1}^n (x_i - \hat{\mu}) = 0$$

$$\Rightarrow \sum x_i - n\hat{\mu} = 0$$

$$\Rightarrow \boxed{\hat{\mu} = \frac{\sum x_i}{n}}$$

Now diff wrt to σ or σ^2 ,

$$\frac{\partial}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum (x_i - \mu)^2$$

$$\Rightarrow \cancel{\frac{1}{\sigma^3}} \frac{1}{\hat{\sigma}^3} (\sum (x_i - \hat{\mu})^2) = \frac{n}{\hat{\sigma}}$$

$$\Rightarrow \sum (x_i - \hat{\mu})^2 = n\hat{\sigma}^2$$

$$\Rightarrow \hat{\sigma}^2 = \boxed{\frac{\sum (x_i - \hat{\mu})^2}{n}}$$

$$\Rightarrow \boxed{s^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2} \quad , \text{ since } \hat{\mu} = \bar{x}$$

Ex: 7.2f (pg 17)

Taking the natural logs of the samples:

→ 0.788, 1.22, 0.470, -0.223, 0.993, 1.194,

0.470, 1.030, 0.916, 0.642

$$\hat{\mu} = \bar{x} = 0.7504$$

$$s_x^2 = 0.4351$$

$$X \sim N(0.7504, 0.4351)$$

$$P[2 < X < 3] = P[0.693 < \overset{\log X}{X} < 1.099]$$

$$= P[-0.131 < Z < 0.8]$$

$$= \Phi(0.8) - \Phi(-0.131)$$

$$= \Phi(0.8) + \cancel{\Phi(-0.13)} - (1 - \Phi(0.131))$$

$$= 0.3405$$

MLE of a Uniform Distribution

$$X_i \sim \text{Uniform}(0, \theta)$$

$$f(x_i) = \begin{cases} 1/\theta, & 0 < x_i < \theta \\ 0, & \text{o/t} \end{cases}$$

$$L(\theta) = \frac{1}{\theta^n} \quad \text{where } \theta \geq \max(x_1, \dots, x_n)$$

As θ increases $L(\theta)$ decreases. So minimizing θ maximises $L(\theta)$ so to be

$$\hat{\theta} = \max(x_1, \dots, x_n)$$

$$\hat{\mu} = \frac{\hat{\theta}}{2}$$

Chapter 7b

Confidence Interval :

Point estimates give a single value for an unknown parameter. CI gives a range of plausible values with an associated confidence level. Thus a ~~high~~ larger interval yields greater confidence. For ex: for a 95% CI procedure, in the long run approx 95% of the intervals constructed this way will contain the true parameter and approx. 5% will miss it.

$\alpha \rightarrow$ proportion of intervals that miss the true param.

$\therefore 1-\alpha = \text{confidence level}$

Let, $X = [X_1, X_2, \dots, X_n]^T$ be a random sample, with θ as unknown param. The CI is written as:

$$(L(X), U(X)) \text{ , then } P[L(X) < \theta < U(X)] = 1-\alpha$$

$\uparrow \quad \quad \uparrow$
~~LB~~ LB UB

from Ch-6 : $E[\bar{X}] = \mu$
 $SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ } we'll use these and CI's to build a range around $\bar{X}$.

when σ is known,

$$\bar{X} - Z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

↳ sample estimate $\pm$ allowance for sampling uncertainty

Factors affecting CI:

- larger sample $\rightarrow$ narrower CI
- higher confidence level $\rightarrow$ wider CI
- greater variability in the sample (s^2) $\rightarrow$ wider CI

Deriving the CI:

Basic form: Estimate $\pm$ margin of error

$$\bar{X} \pm \underset{\substack{\uparrow \\ \text{critical} \\ \text{value}}}{Z_{\alpha/2}} \frac{\sigma}{\sqrt{n}}$$

For a 95% CI, $\alpha = 0.05$

$\frac{\alpha}{2} = 0.025 \rightarrow$ equally distributed between left and right tails.

We need a Z value

that leaves 2.5% in the right tail

$$\rightarrow P[Z > Z_{0.025}] = 0.025$$

$$Z_{0.025} = 1.96$$

CJ Table (Mem.)

Confidence	α	Each tail ($\alpha/2$)	Critical value (z)
90 %	0.1	0.05	1.65
95 %	0.05	0.025	1.96
99 %	0.01	0.005	2.58
99% Upper CI	0.05 0.01	x	2.33

One Sided Intervals

Account for only one tail

- left tail → sets LB
- right tail → sets UB.

• $(L, \infty) : \mu > L$ (upper CI)

• $(-\infty, U) : \mu < U$ (lower CI)

Ex 7.3d (pg 29)

$$W \sim \text{Normal}(\hat{\mu}, 0.3)$$

$$\hat{\mu} - 0.1 < \hat{\mu} < \hat{\mu} + 0.1$$

$$Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}} = 0.1$$

$$\Rightarrow 1.96 \times \frac{0.3}{\sqrt{n}} = 0.1$$

$$\Rightarrow n = 34.57$$

$$\underline{\underline{n \approx 35}}$$

T-Distribution:

Used when σ is unknown, so we use S (sample sd) which ~~accounts~~ adds extra uncertainty. T-distribution accounts for this added uncertainty.

$$P\left[-t_{\alpha/2, n-1} < \frac{\bar{X} - \mu}{S/\sqrt{n}} < t_{\alpha/2, n-1}\right]$$

$$= \bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$$

$$\boxed{CI = \bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}}$$

CI for Variance of a Population:

Suppose $X_1, \dots, X_n$ is a random sample where $X_i \sim N(\mu, \sigma^2)$ where σ^2 is unknown.

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{x})^2}{n-1}$$

and $\frac{(n-1)s^2}{\sigma^2}$ has a Chi-square distribution.

$$P\left[\chi^2_{1-\alpha/2, n-1} < \frac{(n-1)s^2}{\sigma^2} < \chi^2_{\alpha/2, n-1} \right] = 1-\alpha$$

Then solve for σ^2 by taking reciprocals which reverses the inequalities.

$$\frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}}$$

$$\boxed{CI = \left(\frac{(n-1)s^2}{\chi^2_{\alpha/2, n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\alpha/2, n-1}} \right)}$$

Ex: 7.4a

Q: Find 95% CI for the diff. between the two population mean failure voltages.

Given $\sigma_A^2 = 40$

$$\sigma_B^2 = 100$$

Let $W = \bar{X} - \bar{Y}$

$$\bar{X} = \frac{1}{14} \sum_{i=1}^{14} X_i \quad \bar{X} \sim N(\mu_1, \frac{40}{14})$$

$$\bar{Y} = \frac{1}{12} \sum_{i=1}^{12} Y_i \quad \bar{Y} \sim N(\mu_2, \frac{100}{12})$$

$$E[W] = E[\bar{X} - \bar{Y}]$$

$$= E[\bar{X}] - E[\bar{Y}]$$

$$= \mu_1 - \mu_2$$

$$\text{Var}(W) = \text{Var}(\bar{X} - \bar{Y})$$

$$= \text{Var}(\bar{X}) + \text{Var}(\bar{Y})$$

$$= \frac{40}{14} + \frac{100}{12}$$

$$= 11.19$$

$$\cancel{W} \sim N(\mu_1 - \mu_2, 11.19)$$

$$\bar{x} = 42.79$$

$$\bar{y} = 55.83$$

$$w = \bar{x} - \bar{y} = -13.048$$

$$\alpha = 0.05$$

$$\frac{\alpha}{2} = 0.025$$

$$z_{0.025} = 1.96$$

$$P[-1.96 < Z < 1.96] = 0.95$$

$$P\left[-1.96 < \frac{W - (\mu_1 - \mu_2)}{3.3452} < 1.96\right] = 0.95$$

$$\Rightarrow -6.5566 < W - (\mu_1 - \mu_2) < 6.5566$$

$$= -6.5566 - W < -(\mu_1 - \mu_2) < 6.5566 - W$$

$$= -6.5566 - (-13.048) < -(\mu_1 - \mu_2) < 6.5566 - (-13.048)$$

$$= (-13.048) - 6.5566 < \mu_1 - \mu_2 < 6.5566 + (-13.048)$$

$$= -19.6042 < \mu_1 - \mu_2 < -6.4916$$

$$CI = (-19.604, -6.491) \quad (A_m)$$